

Jannatul Ferdous Srabonee

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[Google Scholar](#) | [LinkedIn](#) | [Portfolio](#)

Merced, California, USA

Research Interests: Human-Computer Interactions, Applied Machine Learning, Data Science, Brain-Computer Interfaces

EDUCATION

PhD Candidate in Electrical Engineering & Computer Science

Aug 2023 - Present

University of California, Merced (UC Merced)

Merced, CA, USA

Major: Human-Computer Interactions

CGPA: 3.78/4.0

Advisor: Ahmed Arif

Dissertation: Novel Input and Interaction Techniques for Motor and Vision Impaired Individuals

BSc in Computer Science & Engineering

Dec 2015 – Mar 2020

Khulna University of Engineering & Technology (KUET)

Khulna, Bangladesh

Advisor: Muhammad Aminul Haque Akhand

RESEARCH EXPERIENCE

Graduate Student Researcher

Aug 2023 – Present

Inclusive Interaction Lab, UC Merced

Merced, CA, USA

Research Engineer

Feb 2022 – Jul 2023

Advanced Intelligent Multidisciplinary Systems Laboratory

Dhaka, Bangladesh

Machine Learning Engineer Intern

Jul 2021 – Dec 2021

Robotech Valley

Dhaka, Bangladesh

Undergraduate Researcher

Jan 2019 – Mar 2020

Intelligence Systems Lab

Khulna, Bangladesh

PUBLICATIONS

Conference Papers

[C.3] BezelBraille: A Braille-Inspired Bezel-Guided Virtual Keyboard for Low Vision and Blind Users

Isna Alfi Bustoni, **Jannatul Ferdous Srabonee**, Gulnar Rakhmetulla, & Ahmed Arif

MobileHCI 2026 [[Paper](#)]

[C.2] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription

Jannatul Ferdous Srabonee, Ohoud Mosa Alharbi, Wolfgang Stuerzlinger, & Ahmed Arif

CHI 2025 [[Paper](#)] [[Presentation](#)]

[C.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data

Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique

IEEE TENSYP 2020. [[Paper](#)]

Journal Articles

- [J.2] Computer Vision-Based IoT Architecture for Post COVID-19 Preventive Measures
Ahsanul Akib, Kamruddin Nur, Suman Saha, **Jannatul Ferdous Srabonee**, M. F. Mridha
Journal of Advances in Information Technology (**JAIT**) **2023**. [[Article](#)]
- [J.1] Autism Detection from 2D Transformed EEG Signal using Convolutional Neural Network
Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique
Journal of Computer Science (**JCS**) **2022**. [[Article](#)]

Book Chapter

- [B.1] Alcoholism Detection from 2D Transformed EEG Signal
Jannatul Ferdous Srabonee, Zahrul Jannat Peya, MAH Akhand, Nazmul Siddique
International Joint Conference on Advances in Computational Intelligence (**IJCACI**), **Springer**
2020. [[Chapter](#)]

POSTERS & PRESENTATIONS

Posters

- [Po.2] AI in Accessibility, Research Week, University of California, Merced. **2026**.
- [Po.1] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy. **2022**. (Virtual)

Presentations

- [Pr.8] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription, Yokohama, Japan, **CHI 2025**.
- [Pr.7] A Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, Bangladesh Association of Software Information Services National ICT Awards, Dhaka, Bangladesh, **BASIS 2022**.
- [Pr.6] Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, International Conference on 4th Industrial Revolution for the Emerging Future, Dhaka, Bangladesh, **4IR 2022**.
- [Pr.5] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy, **2022**. (Virtual)
- [Pr.4] COVID-19 Defense System, **IEEE YESIST12** Maker Fair, Dhaka, Bangladesh, **2021**.
- [Pr.3] Mobile App for Post COVID-19 Measures, NASA Space Apps Challenge, Dhaka, Bangladesh, **2021**.
- [Pr.2] Alcoholism Detection from 2D Transformed EEG Signal, International Joint Conference on Advances in Computational Intelligence, Daffodil International University, Dhaka, Bangladesh, **IJCACI 2020**.
- [Pr.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data, Dhaka, Bangladesh, **IEEE TENSYP2020**.

TEACHING EXPERIENCE

Teaching Assistant, UC Merced

Aug 2023 – May 2026

Algorithm Design and Analysis (Fall 23), Introduction to Human Computer Interaction (Spring 24), Introduction to Human Computer Interaction (Fall 24), Introduction to Programming (Spring 25), Introduction to Programming (Spring 26), Discrete Mathematics (Spring 26)

CORE SKILLS

Languages: Python, C/C++, MATLAB, JavaScript, HTML, CSS

AI/ML Frameworks: TensorFlow, Keras, scikit-learn, OpenCV

Data Manipulation & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Research Methods: User Studies, Usability Testing, Experimental Design, Mixed-Methods Research, Interviews, Focus Groups, Quantitative Data Analysis, Statistical Hypothesis Testing

FELLOWSHIPS & AWARD

- EECS Bobcat Fellowship, (2024, 2025, 2026), University of California, Merced, CA, USA
- Winner Award, 2022, Bangladesh Association of Software Information Services (BASIS) National ICT Awards, Dhaka, Bangladesh
- General Scholarship in Higher Secondary School Certificate examination, 2015, Board of Intermediate and Secondary Education, Dhaka, Bangladesh

ACADEMIC SERVICES

Student Volunteer

CHI 2025

Apr 2025 - May 2025

Yokohama, Japan

Senior Executive Committee Member

IEEE Student Branch KUET

Feb 2019 - Feb 2020

Khulna, Bangladesh

PROFESSIONAL AFFILIATIONS

Association for Computing Machinery (ACM)

since Sep 2023

Health Sciences and Research Institute (HSRI) at UC Merced

since Nov 2024

PROFESSIONAL EXPERIENCE

Python Developer

Mir Info Systems Ltd

Jun 2022 – Oct 2022

Dhaka, Bangladesh